

# Urbanization Meets the Environment

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## Abstract

Climate-related disasters such as floods are increasingly severe due to climate change. This is particularly the case in cities witnessing rapid urbanization and widespread poverty. In these settings, mobilizing local communities for flood prevention may be key to economic development. This paper follows a community-level intervention in a Mozambican coastal city with increasing population pressures and recurrent floods. It nudged communities and leaders to select and implement their own climate adaptation projects. This was achieved through the organization of a sequence of community meetings and the provision of technical information, validated by local and international water experts and made specific to the circumstances of each block of the city. The technical information included details of suitable small-scale infrastructural improvements, e.g., drainage canals, as well as of appropriate waste management. The intervention also offered seed grants to help local resource mobilization. We employ a clustered randomized controlled trial design with multiple survey rounds, supplemented by data from structured community activities, enumerator observations, photographs, and focus group discussions. Our results during/after the first rainy season following the intervention show that the treatment shifted attitudes toward personal responsibility and collective efficacy of local block leaders, who are seen as central to enabling impact, and improved observed localized cleanliness. We also register less flooding in flood-prone points and increased walkability. We conclude that a community-driven approach to disaster risk prevention can be effective. These are preliminary results based on short-term outcomes; a further round of endline data collection is pending.

JEL Classification: C93, D90, H40, O18, Q54

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# 1. Introduction

The effects of climate-related disasters are exacerbated in cities due to rapid urbanization and increasing population density, primarily when poorly planned and occurring in a setting of widespread poverty. Mozambique's coastal cities are frequently hit by extreme weather events, like Cyclone Idai, which hit the city of Beira in Sofala province, leading to 600 deaths and affecting more than 1.5 million people in 2019. Mozambique's coast is one of the world's hotspots for extreme weather events: it has a long history of catastrophic flooding, primarily influenced by La Niña and the Intertropical Convergence Zone.<sup>1</sup>

At the same time, urbanization processes are happening across Africa, often quickly and unregulated. Due to conflict and climate change, both rural economic migrants and internally displaced individuals see coming to the city as an opportunity to start again and improve their lives. In this context, constructing new informal houses in hazard-prone and risky areas is frequent. This is all combined with and exacerbated by changes to the built environment at the neighborhood level. Mangroves, for example, which are crucial for protecting against floods and erosion, are being cut down at an unprecedented rate to create space and building materials. These localized changes can have compounding consequences on flooding and hazards.

In this project, we ask whether providing technical assistance and enhancing community cooperation can help control environmental and disaster risks, such as flooding in the coastal city of Quelimane. We are specifically interested in testing the impact of providing information about locally viable adaptation projects for flood mitigation, complemented by a small grant and meetings to promote cooperation for their successful implementation. Hence, our intervention eases technical capacity, liquidity, and cooperation constraints. Our innovative approach to community-driven disaster risk management aims to facilitate the implementation of a climate adaptation project chosen by the community itself.

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1. See also the [World Bank Climate Change Knowledge Portal](#).

In the first part of the analysis, we study the mechanisms underlying the intervention and test hypotheses on its effects for outcome families with the data already collected (additional outcomes and overall effectiveness will be assessed after endline data collection). We focus on the intervention’s short-term impact on community leaders’ locus of control, behaviors, and efforts; on community mobilization and cohesion; and on community members’ collective and individual preventive behaviors. To this end, we draw on three sources of data collected immediately after the intervention but before the rainy season: (i) a structured community meeting organized by block leaders, (ii) a brief midline leader survey, and (iii) systematic photographs of the drainage system. The drainage photographs were repeated during the rainy season to track preventive behaviors. We also assess the intervention’s effects on flooding and its consequences for community livelihoods, using flooding and walkability data collected during the rainy season. Finally, we report findings from focus group discussions that provide qualitative insights into the intervention’s dynamics.

We find strong community mobilization in treated blocks: when the project was implemented, 4,156 individuals participated as volunteers. The intervention had modest but meaningful effects on flood-related behaviors and outcomes. Responses from a block leader survey indicate a shift in leader attitudes toward personal responsibility and collective efficacy in disaster preparedness, an interesting result, particularly given that they were described as central to enabling impact in focus group discussions. Treated communities reported higher levels of localized cleaning, especially in areas close to baseline household houses. Trash detection using computer vision showed no consistent reductions, but enumerator-rated cleanliness improved in treated blocks. We also detect less frequent flooding and greater ease of mobility in treated areas, suggesting improved resilience to seasonal flooding. At a more aggregated level, drainage canal segments and block centers in treated communities showed increased evidence of drainage connectivity and community-led cleaning.

Beyond these behavioral mechanisms, the project also generates rich spatial

data that allows us to study how risks, responses, and adaptation strategies vary across neighborhoods. In this paper, we focus on the cross-sectional baseline survey of local chiefs and residents to examine behaviors around flooding and willingness to pay for disaster insurance. Spatial correlations and exploratory analyses reveal patterns and hotspots across the city where individuals and neighborhoods share similar flooding experiences, preferences, and mitigation strategies, and we further explore how these outcomes differ in higher-risk areas, such as those closer to the coastline or at lower elevations.

These spatial analyses are supported by a wide range of geographic information generated by technical experts, local communities, and field enumerators, complemented with publicly available land-use and population data. Maps and visual graphics were used extensively to present information in the field and to visualize the context and project interventions, including updated data on drainage systems and their typology, mangrove planting areas, observable water amenities, and coastal exposure, as well as block-level flood risk.

These analyses connect to a broader literature. Bryan et al. (2020) argue that coastal cities are particularly vulnerable in a climate scenario of increasing extreme weather events. Nevertheless, the literature on the intersection of urbanization and climate adaptation still needs to be explored, especially in developing settings. The related literature comes from a diverse set of scientific fields. We highlight four key areas.

First, we build upon prior research led by our team, which identified tangible community responses to flood mitigation measures in Quelimane, particularly in waste management strategies (Leeffers, 2024). This work extends to understanding the critical role of cities' public policies in integrating rural migrants (Armand et al., 2022).

Second, we draw from evidence that underscores the importance of thoughtful urban planning and infrastructure investment as pivotal to reducing flood risks (Gandhi et al., 2022) and emphasize the necessity of financial and technical support to facilitate such endeavors (Lumbroso et al., 2008; Abagat et al., 2017).



Third, we acknowledge the significant role of local governance in adaptation strategies (Agrawal et al., 2008) and the success of community-based organizations in mitigating flood risks, as seen in Senegal (Newman et al., 2024). Our proposal ventures to answer how best to incentivize and foster the establishment of such community organizations.

Fourth, as flood-prevention infrastructure is a public service, and forming committees and holding meetings could induce community-based monitoring, we build on previous studies that investigate the impact of community-based monitoring on public service delivery. In the Ugandan health sector, Björkman and Svensson find that a community-driven local accountability project resulted in significant improvements in health care delivery, utilization, and health outcomes after one year (Björkman and Svensson, 2009) and in the longer run (Björkman Nyqvist et al., 2017). However, Raffler et al. (2020) test a related intervention in a similar setting and show that it does not increase community monitoring, nor generate improvements in health outcomes. Our study contributes to this literature by adding evidence on the effects of community-based monitoring on infrastructure as a public service.

## 2. Experimental Design

### 2.1. Intervention

To address three critical constraints to climate adaptation infrastructure — technical knowledge, liquidity, and cooperation — we designed a multifaceted intervention, implemented between April 25, 2024, and December 11, 2024. The intervention tackled each constraint through a distinct but interconnected component: (i) engineering-backed technical assistance, (ii) seed grants allocated at the block level, and (iii) a sequence of participatory meetings. These components were designed to be both independently impactful and mutually reinforcing.

### *2.1.1. Technical Assistance*

To address technical knowledge gaps, we proposed a flood mitigation project for each city block. To ensure feasibility and rigor, we established partnerships with international and local experts. Together with the Portuguese university Instituto Superior Técnico and the Swiss engineering firm MHYD, we collected in situ data, conducted hydraulic and hydrological studies, and developed a macro water model for Quelimane. We mapped the drainage canal system, identified key flood risk areas and zones suitable for mangrove replanting, and produced detailed maps to guide implementation. The international engineering team recommended a set of flood prevention measures, which formed the foundation for block-level projects. Local engineers validated these proposals for financial and logistical feasibility, incorporating insights on capacity and resource availability. To further strengthen the mangrove replanting component, we partnered with the Centro de Pesquisa e Tecnologia do Mar (CePTMar) and two Quelimane-based restoration associations, ANAMA and ASSOPEZA. These collaborations ensured that the proposed measures were technically robust and locally appropriate.

### *2.1.2. Seed Grants*

To address liquidity constraints, we provided each city block with a small grant to support project implementation. Valued at 5,000 MZN (approximately 80 USD), the grant was earmarked for purchasing materials. It was not distributed directly to community members.

### *2.1.3. Participatory Meetings*

To promote cooperation, we organized three rounds of participatory community meetings. Block leaders were encouraged to invite all residents to participate. Our team supported attendance through door-to-door mobilization and SMS reminders.

**First meeting: Identifying challenges** In the first meeting, communities discussed flood-related challenges, identified local issues, and proposed potential solutions. Participants categorized these into smaller, locally manageable problems

and larger, systemic issues. They then prioritized the former. Our team committed to conducting a technical evaluation of the top-ranked suggestions before the second meeting. Each block also formed a local committee — comprising the block leader and elected residents — to facilitate decision-making. Finally, we introduced the grant and explained its purpose and disbursement process.

**Analysis between meetings** After the first round of meetings, we reviewed the proposed measures from a technical perspective. This process yielded five types of technically feasible and community-suggested flood mitigation projects: 1) block and/or drainage canal cleaning; 2) opening and connecting canals to the system; 3) *Entulho* (filling holes with debris to reduce water accumulation and improve access) or *Abaulamento* (elevating roads to improve drainage); 4) replanting mangroves; and 5) constructing or repairing bridges.

We developed an algorithm to recommend a project for each block. It incorporated: a) community suggestions, b) geographic and environmental block characteristics, and c) expert data, including the validated flood prevention measures, drainage canal maps, and mangrove suitability zones. This algorithm ensured that each recommended project reflected both local priorities and technical best practices. It offered a scalable model for participatory infrastructure planning.

**Second meeting: Finalizing projects** In the second meeting, we presented each project as a recommendation and encouraged the community to make the final decision. This approach was designed to respect local knowledge and foster a sense of ownership. To support informed discussion, we shared detailed block-level maps of the drainage network and mangrove zones. Projects were finalized through consensus between community input and technical guidance.

After agreeing on a project, participants identified the necessary materials and volunteer labor. We reiterated the grant’s role and encouraged residents to contribute time or resources, reinforcing shared responsibility.

**Third meeting: Implementing projects** The third meeting focused on implementation. Guided by the local committee and supported by our team, residents completed the agreed-upon project using the allocated materials and volunteer labor.<sup>2</sup>

Alongside the primary project implementation, we rolled out an awareness campaign to encourage positive behaviors and discourage harmful ones. Many communities had proposed this as a cost-effective complementary activity during the first meeting.

We developed the campaign content based on focus group discussions with a block leader, an ASSOPEZA member, and other stakeholders. The campaign had two phases: 1) megaphone messaging, with announcements in Portuguese and Chuwabu to ensure broad comprehension; and 2) brief community lectures reinforcing key messages and encouraging continued engagement.<sup>3</sup>

At the end of the lectures, we distributed two maps to reinforce awareness: 1) a drainage map showing canal typologies in the block and adjacent areas; and 2) a flood risk map with block-specific exposure levels. For mangrove-replanting blocks, we also distributed a one-page planting guide developed in collaboration with CePTMar, ANAMA, and ASSOPEZA.

Finally, to reinforce the campaign, we sent follow-up messages via SMS to all contacts gathered during the door-to-door mobilization and meeting phases.

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2. Our team purchased the materials, and block leaders collected them at a designated location and time using vouchers. For *Entulho* projects, sand was delivered directly by truck. For drainage cleaning projects, waste removal was also coordinated. Materials remained with the community after the intervention, with committees responsible for storage and reuse.

3. The megaphone messages and community lectures cover a range of critical topics, including: a) recommendations for effective waste management; b) advice against construction on drainage canals or wetland areas; c) guidance on implementing *Entulho* or *Abaulamento* with suitable materials; d) guidance on safe bridge construction; e) promotion of mangrove preservation and discouragement of mangrove cutting; f) encouragement of planting vegetation around homes; g) encouragement of communication within communities about weather alerts; h) promotion of community collaboration in flood prevention efforts.

## 2.2. Sampling

There are 557 blocks in 54 neighborhoods in 7 administrative posts in Quelimane. Building on GIS data for the Quelimane basin area and exploratory surveys with local leaders, we selected 489 urban and peri-urban blocks in the city of Quelimane. In total, 68 blocks were excluded from the sample, namely the blocks within Administrative Post 5 (due to its rurality and low population density) and the downtown blocks (due to the developed drainage system in this area).

To sample households, six random houses within each city block were pre-selected and ranked in QGIS. The random locations were divided into male and female respondents to enable a gender-balanced household sample. Enumerators were instructed to visit two locations per gender and interview the household head (or partner) of the assigned gender. Interviews with unavailable respondents were rescheduled for a later date.<sup>4</sup>

For the block leader sample, one leader per city block was surveyed. If the block leader was unavailable for the entire duration of the baseline or if the block had no assigned leader at the time, the enumerator asked the neighborhood secretary to name the second-best person to answer our survey. This person was requested to have a block leadership position or to be a long-time resident.

## 2.3. Clustering

The blocks are categorized into eleven drainage zones based on watersheds and drainage systems. Within each zone, a clustering algorithm is applied utilizing the heuristic approach outlined by Wei et al. (2020), implemented through the Max-P Regionalization Algorithm from the PySAL Library.

This clustering process involves setting a minimum of three adjacent blocks and considering attribute variables such as altitude and type of drainage canal. Subsequently, we refine these clusters in a post-processing phase, prioritizing

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4. If any of the households were ineligible (non-residential location, non-adult household head, gender assigned not applicable), the enumerators would visit the third randomly selected location. If more substitution was needed, the enumerators were instructed to visit the closest house to the yet unused location until they had surveyed two male and two female respondents per city block.

allocations consisting of three blocks and promoting increased border sharing and drainage canal connectivity. Approximately 16% of the blocks are adjusted during this phase, resulting in the definition of 159 distinct clusters.

## **2.4. Randomization**

We randomly allocated treatment within the 159 clusters stratified by drainage zone and the number of blocks with drainage canals. This results in 80 clusters allocated to treatment, comprising 246 blocks. The resulting allocation of clusters, blocks, and households to treatment and control can be found in Table 1. See Figure 1 for an overview of treatment distribution. Note that randomization was performed at the cluster level to make sense of flood-prevention projects with clear geographical spillovers (e.g., block-level improvements in the drainage system). At the same time, treatment was administered at the block level to leverage community cohesion and trust.

## **2.5. Theory of Change and Hypotheses**

We provide a theory of change in Figure 2. Beyond measuring the overall effectiveness of the intervention, we will investigate underlying mechanisms and test a range of hypotheses regarding its effects on immediate outcomes (community members' demand for leader accountability; knowledge and awareness among community members and leaders; community mobilization and cohesion; locus of control among community members and leaders), intermediate outcomes (community leaders' behaviors and efforts; their collective and individual preventive behaviors regarding floods; and their high-risk behaviors regarding floods), and final outcomes (flooding; the impact of flooding on community members' livelihoods). Figure 2 also describes the links between the various outcomes.

In this paper, we focus only on outcome families for which data have already been collected: community mobilization and cohesion (structured community activity); leaders' locus of control (short leader survey); community leaders' behaviors and efforts (structured community activity and short leader survey); and environmental

risk mitigation and exposure. The latter combines two data sources: (i) systematic photographs and structured field observations collected in December 2024 and March 2025 along drainage canals and in the center of blocks; and (ii) direct walkability and flooding observations recorded in March 2025 along predetermined routes through each block.

### 3. Empirical Strategy

#### 3.1. Estimation of Treatment Effects

The impact of the intervention will be assessed by comparing outcome variable means between treatment and control groups, using this regression specification:

$$y_{i,b} = \alpha + \beta_1 T_b + \delta_1 \mathbf{X}_{i,b} + \theta_1 \mathbf{Z}_b + \omega_1 y_{0,i,b} + \varepsilon_{i,b} \quad (1)$$

where  $y_{i,b}$  is the outcome of interest for respondent or location  $i$  in block  $b$  at endline.  $y_{0,i,b}$  is the corresponding outcome at baseline.<sup>5</sup>  $T_b$  is a binary variable taking the value 1 if the block is assigned to the treatment.  $\mathbf{X}_{i,b}$  is a vector of individual control variables, including the respondent's gender, age, education, and household size.  $\mathbf{Z}_b$  is a vector of block-level control variables, including the number of houses, the area size in hectares, meters of drainage canals, average elevation in meters, neighborhood, and strata fixed effects. Finally,  $\varepsilon_{i,b}$  is the usual idiosyncratic error term. The coefficient  $\beta_1$  is our estimated treatment effect. Standard errors are clustered at the cluster level.

To ensure our analysis is not distorted by variables that show very little change, we exclude any variable where over 95% of the observations within the relevant sample are identical. Additionally, responses of "do not know" or "refuse to answer" are treated as missing data — we do not attempt to fill in these missing values.

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5. We only control for baseline values to increase power in this ANCOVA specification if we have the corresponding variable (or all variables that constitute an index) at baseline. Some questions are added at the endline only, so we will not be able to control for the corresponding outcome variable at baseline.

For continuous variables, trimmed values are used to reduce the influence of outliers. In particular, we trim 2.5% of each side of the distribution for household-level outcomes and 1% of each side of the distribution for block-level outcomes.

Inverse hyperbolic sine (IHS) transformations are used if variables are skewed, with skewness being defined as the adjusted Fisher-Pearson coefficient of skewness exceeding 1.96. To address the recent critiques by [Mullahy and Norton \(2022\)](#) and [Chen and Roth \(2024\)](#), we will apply the IHS transformation cautiously when dealing with outcome variables that have many zeros and impacts on the extensive margin. Specifically, we will estimate and document treatment effects on the extensive margin, note the proportion of zeros in the data, carefully evaluate how much we value extensive versus intensive margin effects in our estimations, and exercise caution in interpreting the magnitudes of treatment effects on IHS-transformed outcome variables as percentage changes.

### 3.2. Spatial Analyses

Since we collect underlying location (GPS) information on chief and household residences, we run a series of spatial correlations and statistics on responses. We focus on baseline survey data where participants (both chiefs and households) were asked similar questions on willingness to pay for flood mitigation, past flood occurrences, and thoughts on the likelihood of damages from future floods.

We calculate a Moran I statistic across the reported willingness to pay for flood mitigation (respondents were asked to contribute values on an ordinal scale from 0 to 10). This global measure of spatial autocorrelation measures a variable's correlation with the neighboring values of itself, providing an overall indicator of how strongly clustered values are over space. From this, we are able to comment on whether neighborhood similarities emerge in how much residents are willing to pay for local flood mitigation. The Moran statistic is given by the formula:

$$I = \frac{N}{\sum_{i=1}^N \sum_{j=1}^N w_{ij}} \frac{\sum_{i=1}^N \sum_{j=1}^N w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$



Here, the autocorrelation between a variable  $x$  with  $N$  observations is calculated. One of the core aspects of this calculation, which differentiates it from a traditional correlation, is the use of a  $N \times N$  spatial weight matrix  $W$ , where each element  $w_{ij}$  measures the strength of proximity between observation  $x_i$  and  $x_j$ . We measure that here by using the inverse Euclidean distance. That is, the strength of each pair-wise relationship over space is inversely proportional to how close these observations are to each other, and the  $x$  values are weighted proportionately.

We then split our dataset into different subsets, which allows us to estimate and compare the values and strength of the relationship for our willingness to pay across the city. Using the spatial nature of the data, we explore how factors such as proximity to the waterfront or living in low-lying areas might yield different patterns and strengths in willingness to pay.

## 4. Data

Several rounds of household and block leader surveys serve as an important data source. To broaden our pool of objective information, enumerators conducted inspections and noted down observations. These are supplemented by data from a set of structured community activities (Casey et al., 2012; Armand et al., 2020), systematic photographs of the drainage system, and focus group discussions.

### 4.1. Surveys

During the baseline survey, implemented between November and December 2023, the field team interviewed four households and one leader per city block, resulting in 1,956 household surveys and 489 block leader surveys.<sup>6</sup> Additionally, enumerators note some observations on the individual and block level. The sample sizes are sufficient to reliably detect meaningful treatment effects for outcome variables at both the household and block leader levels; details of the power calculations are

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6. High-frequency checks were performed daily to ensure data quality and protocol compliance. Moreover, control calls and/or back-checks were conducted on 10% of the sample, where the enumerator would ask a random set of 5-10 questions from the baseline survey.

provided in Appendix A.1.

Importantly, block leaders responded to five questions from the baseline questionnaire in December 2025, shortly after the intervention and before the rainy season. This data collection, referred to as the *short midline leader survey*, provides outcome variables for which treatment effects are reported in Subsubsection 5.3.2.

#### 4.1.1. Descriptive Statistics

Descriptive statistics can be found in Table 2. Our sample of 1,956 households comprises families with 5.4 members on average. Respondents, either the household head or their partner, are on average 39 years old. Most have completed primary school (72%), while 40% have completed secondary education. About 68% report having worked in the last 12 months, and the average monthly household income is roughly 9,366 MZN (around 150 USD), though more than half live on less than 5,000 MZN (around 80 USD). Indicators of vulnerability are pronounced: 65% of households report unmet food needs, 42% report living with difficulties, and the average household has experienced nearly two flooding episodes in the past.

Block leaders are older on average (50 years) and have held their position for over five years. 33% are female, and 75% were employed during the last year. They report an average of three flooding episodes in their blocks, with 10% reporting seven or more.

At the block level, 67% of the 489 blocks have drainage canals, though most are uncovered and unpaved. The average block contains about 192 buildings, spans nearly 6.8 hectares, and is located at an elevation of about 7.7 meters. Only 3% of blocks had no leader appointed at the time of the survey.

#### 4.1.2. Randomization Balance

To assess whether treatment and control groups are comparable at baseline, we test for differences in the average values of household, block leader, and block characteristics. The set of variables includes both outcomes unlikely to be affected by the intervention and those later used to measure its impacts and underlying mechanisms.

On the majority of baseline characteristics, the sample is balanced between treatment and control groups. Tables 3 and 4 present the full set of balance checks across households, block leaders, and block characteristics, with all comparisons accounting for strata fixed effects. Differences between treatment and control groups are minimal: among 12 household-level comparisons, none is statistically significant at the 10% level. At the block level, only one of 19 comparisons is statistically significant at the 10% level. Overall, these results indicate that treatment and control clusters are balanced on observables at baseline.

#### 4.2. Structured Community Activity 1: Community Gathering

We conduct structured community activities following Casey et al. (2012) and Armand et al. (2020). In the first one, implemented December 15-20, 2024, right after the intervention and before the rainy season, we asked block leaders to gather their communities to discuss the challenges that arise with the approaching rainy season and potential preventive measures. The leader was responsible for contacting the block members and mobilizing them to participate in the gathering.

We provided leaders with an incentive to support the organization of the gathering. We also asked them to note on paper whom they have contacted to attend the gathering (names, phone numbers, and the way they have contacted them).

On the day of the gathering, enumerators visited the blocks and recorded the number of community members who were present, the number of community members who spoke during the meeting, and the topics covered in the discussion. They also collected the paper with the community members who have been contacted by the block leader to attend the gathering.

This structured community activity measures community mobilization and cohesion on the one hand and leader effort and behavior on the other hand.

### 4.3. Systematic Photographs and Direct Field Observations

To capture both the community-level responses to environmental risks and the presence of flooding, we collected systematic photographs and structured field observations across all study blocks. These data were gathered in two rounds, December 2024 (shortly after the intervention and before the rainy season) and March 2025 (during the peak of the rainy season), across multiple block locations and observation types.

Enumerators visited three types of locations: (i) every 50 meters along drainage canals, (ii) at central points within each block, and (iii) along fixed walkability routes through each block. At each site, they took systematic photographs from multiple angles and recorded structured observations on cleanliness, flood preparedness, and flooding signs.

**Systematic photographs** The photographs were used to detect the presence of trash. Building on [Leeffers \(2024\)](#), who used a vision-language model to classify images based on natural language queries (e.g., “Is there trash?”), we introduce a refined approach that uses object detection techniques to identify and localize pieces of trash directly. This transition from image classification to object detection allows for a more granular and interpretable measure of cleanliness, addressing limitations in the earlier methodology where trash presence could only be inferred from high-level labels.

Following [Jakob and Coccia \(2024\)](#), we trained a custom object detection model using RetinaNet with a ResNet-50 backbone. We fine-tuned this model on a hybrid dataset combining hand-labeled images from our fieldwork (n=300) and curated images from the publicly available TACO dataset ([Proença and Simões, 2020](#)), filtered for environmental similarity (vegetation, dirt, and water backgrounds). Our combined training set included 670 images, with an additional 200 images used for validation and testing.

Detecting trash in our context presents unique challenges. Most items are small, often occluded, or partially buried in soil and vegetation. The majority of

bounding boxes are under 50 pixels in width and height — making them difficult targets for standard detection algorithms. We therefore modified the anchor sizes in RetinaNet to improve performance on small objects and applied data augmentation techniques, including horizontal flipping and variable scaling, to increase model robustness.

We report an AP50 (average precision at 50% intersection-over-union) of 33.4% and an APs (average precision on small objects) of 18.9%. While these metrics are slightly lower than those achieved on benchmark datasets like COCO, they are highly competitive when restricted to small-object categories. Our model’s performance on small-object detection is thus comparable to benchmark results in similar domains, a notable achievement given the noisier and less controlled photographic conditions of our dataset.

To operationalize cleanliness from the model’s output, we use the number of trash detections (i.e., bounding boxes) identified per image as our primary outcome variable. For each location, we calculate the average number of trash detections across the five images, providing a single, interpretable metric that captures the local intensity of visible waste. An example of the model’s detection output is shown in Figure 3, illustrating the typical distribution of small-scale trash objects that our approach captures.

**Direct field observations** To complement this automated analysis, enumerators visiting each location also recorded structured observations on site cleanliness, recent cleaning activity, other recent preparedness actions, and visible signs of flooding. These human-coded assessments offer a valuable additional lens through which to interpret the local conditions, capturing elements of perceived cleanliness and risk that may be missed or misclassified by machine learning models, particularly in cluttered or ambiguous contexts.

**Flooding and walkability** The fixed flooding and walkability route component was only implemented in March 2025. During this activity, enumerators also assessed flooding at seven locations per block: (a) at one GPS-defined low-lying

point based on elevation maps, (b) in front of four baseline households, (c) at one block border point next to a tarmac road, and (d) at the most visibly flooded location in the block.<sup>7</sup> They recorded flood depth using a measuring stick.

To evaluate the effects of flooding on mobility, enumerators walked three predetermined paths within each block: (i) from the likely flooded point to the first baseline household, (ii) between each of the four baseline households, and (iii) from the last household to the block border point. For each route, they measured walking time, recorded whether the direct path was passable, and noted the cause of any deviations (e.g., standing water, physical barriers). They also rated the ease of movement on a standardized scale. These walkability data help us assess the direct consequences of flooding on daily navigation and accessibility.

Together, the systematic photographs and direct field observations provide a high-resolution, spatially structured dataset that captures both visible environmental risks — such as trash accumulation — and real-time impacts of flooding on block-level mobility and preparedness.

#### 4.4. Focus Group Discussions

We conducted Focus Group Discussions (FGDs) in ten blocks from the treatment group, selected to capture variation in outcomes: six blocks where the intervention was particularly successful and four where it was less so. The classification of blocks as either successful or unsuccessful was made by the local field coordinator, drawing on his observations during implementation as well as residents' feedback at the time. Control blocks were excluded from the qualitative sub-sample, since the purpose was to explore how and why results differed across treatment sites. The selected blocks included all five types of implemented measures: three focused on block or drainage cleaning, one on opening and connecting canals to the drainage system, two on *Entulho* or *Abaulamento*, two on mangrove replanting, and two on constructing or repairing bridges.

FGDs were carried out between July 2 and July 7, 2025, eight to nine months

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7. In blocks where no flooding was observed, enumerators visited six points instead of seven.

after the final stage of the intervention. Each discussion lasted 45 minutes to 1 hour and 20 minutes, with 11 to 36 residents participating. Block leaders assisted with organization and scheduling. The local field coordinator moderated the discussions, while an assistant moderator took notes and produced one summary report per FGD. Most discussions took place in Chuwabu, the local language, which allowed for open and fluid exchanges. The semi-structured guide combined open-ended questions, probes, and follow-ups, covering previous flooding experiences, leadership and group dynamics during the intervention, and residents' perspectives on the implemented measures. Our aim was both to identify what distinguished more successful from less successful blocks and to consider whether alternative organizational approaches could have increased effectiveness.

To avoid influencing endline survey answers, the coordinator provided block leaders with a list of residents not to invite, as they had already participated in the baseline and were scheduled for the endline survey. Block leaders themselves were also excluded. This also served to encourage participants to speak freely, including when criticizing leadership or the organization of activities. All discussions were recorded. Our conclusions draw on the ten written reports. For analysis, we classified information into three categories: 1) problems identified by residents; 2) suggested solutions; and 3) factors or dynamics distinguishing successful from less successful blocks.

## 5. Results

### 5.1. Spatial Analyses

We first present some exploratory spatial analyses to look at measures of flooding preferences and responses across the city. Flooding experience and relationships can be very localized, and we are looking to explore whether there is significant similarity and (spatial) clustering in individuals' willingness to pay for flood mitigation. Globally, the Moran I estimate of 0.025 indicates relatively weak spatial clustering, meaning that across the whole sample, the willingness to pay of

an individual is only slightly correlated to what their neighbors are willing to pay.

Table 5 shows estimates for the Moran I for various subsets of the survey, conditional on location. We use geospatial data and variables to subset the survey responses according to different location characteristics — using median values to split the data into high and low partitions (except for mangrove adjacency, which is defined as being in a block bordering mangroves). The Moran I spatial clustering value is accompanied by the average value for the willingness to pay variable within each subset.

Figure 5 visualizes these impacts by mapping the values for the willingness to pay across survey respondents. We map the value responses for willingness to pay (points) against the background map representing self-reported flood occurrences from respective block chiefs. We can see hotspots across the city and some clustering of values. Concentrations of these values indicate that residents in these areas have similar preferences with respect to their willingness to pay for flood mitigation. Some clear spatial patterns emerge, suggesting the importance of considering location and context in flood management preferences.

While we may observe relatively strong similarities in how people behave across different spaces — that is, specific neighborhoods having similar willingness to pay for flood defense, this may vary in terms of the magnitude of whether individuals are willing to contribute more or less towards flood mitigation. For example, while there is relatively stronger clustering of values for those respondents closer to the coastline, the overall willingness to pay between those near the coastline and those who are not is similar. For respondents who are nearer to mangrove plantations, we see a similar willingness to pay, but overall, it is less in magnitude on average compared to those who are not near mangroves.

## 5.2. Intervention Statistics

**Community-suggested measures** During the first community meeting, 7,347 participants gathered to propose locally manageable, small-scale flood mitigation measures (for a detailed description of the intervention, see Subsection 2.1), with



each block suggesting an average of three flood mitigation interventions.

In particular, they suggested 238 instances (31%) of *Entulho* (filling holes with debris to reduce water accumulation and improve access) and *Abaulamento* (elevating roads to improve drainage), both of which are common practices in Quelimane. However, if carried out without consideration of factors such as the location of natural watercourses, these practices can exacerbate flooding. The prevalence of this suggestion highlights the importance of awareness-raising activities to reduce the risk of unintended negative consequences.

Communities proposed 149 instances (19%) of block and/or drainage canal cleaning, underscoring the pervasive challenge of improper waste disposal across the city. In addition, 84 instances (11%) involved suggestions to open and connect canals and natural water channels to the drainage system, pointing to opportunities for improving the maintenance and expansion of existing drainage infrastructure.

There were 107 instances (14%) in which community members proposed awareness-raising activities as a solution to flood-related challenges. This reflects concern about harmful practices, such as improper waste disposal, and a belief that disseminating information on the associated risks could positively influence neighbors' behavior. Additionally, we recorded 23 instances (3%) of suggestions to construct or repair bridges.

We recorded only 18 instances (2%) of proposals for mangrove replanting to mitigate flood-related issues. This likely reflects the fact that mangrove restoration is feasible only in specific areas of Quelimane, as mangroves require saline water and suitable soils that are absent in many blocks. Furthermore, urban development in numerous neighborhoods limit the potential for mangrove growth. Consequently, replanting efforts should be targeted to coastal or flood-prone zones where mangroves can thrive and provide protection, while blocks outside these areas are unlikely to propose them.

Finally, 149 instances addressed other non-environmental needs, such as improvement of street lighting, drinking water access and means of access.

**Algorithm-suggested measures** We then assessed the proposed measures from a technical perspective. To identify the most suitable recommendation for each block, we developed an algorithm that integrates: (a) the community’s suggestions from the first meeting, (b) the geographic location and specific characteristics of each block, and (c) our engineering expertise, drawing on validated flood-prevention measures, a detailed drainage system map (Figure 4), and a map of areas proposed for mangrove replanting (Figure 6).

This approach generated one financially feasible flood-prevention project recommendation per city block, ensuring that each recommendation aligns with local priorities and engineering best practices while fostering community participation and ownership. Specifically, the algorithm produced the following projects: 221 instances of block or drainage canal cleaning *or* opening and connecting canals to the drainage system (90%), and 25 instances of mangrove replanting (10%).

**Implemented measures** During the second community meeting, our team presented these flood-prevention projects to each block as recommendations rather than directives, ensuring that communities retained the final decision. This approach acknowledged their deep knowledge of local challenges and strengthened a sense of ownership over the chosen projects. Consequently, the final selection in each block emerged from a collaborative process that combined community proposals, our technical assessment, and a final collective discussion during the second meeting.

This process led to the following implemented projects: 113 instances of block and/or drainage canal cleaning (46%), 17 instances of opening and connecting canals to the drainage system (7%), 90 instances of *Entulho* or *Abaulamento* (37%), 12 instances of mangrove replanting (5%), and 13 instances of constructing or repairing bridges (5%). All 246 treated blocks also received the awareness-raising campaign (in addition to one main project that every block agreed upon); one block received only this campaign without any additional project. Figure 7 shows the implemented project distribution on a map.

**Community mobilization** Once a project was agreed upon, the focus during the second meeting shifted to identifying the necessary materials and volunteers. Our team encouraged the 5,171 community members present to contribute their labor and any materials, reinforcing collective effort and ownership. In total, 4,292 individuals volunteered their labor (83% of participants in the second meeting), averaging 18 volunteers per block. 37% of participants pledged material or monetary contributions such as tools or food to support the initiative, averaging seven material contributors per block.

During the third and final meeting, the community gathered to implement the project. On this day, 4,156 individuals were present as volunteers, averaging 17 per block, showcasing their readiness of to support the initiative with physical labor.

Our team procured the required materials, which block leaders collected at a designated location and time. Ninety percent of block leaders collected the materials for implementation on the scheduled day without requiring reminders.

### 5.3. Treatment Effects

We now present the impacts of the intervention on pre-specified outcomes. To ensure transparency and replicability, all outcome variables were registered in a pre-analysis plan, available in the [American Economic Association’s registry for randomized controlled trials](#). Effects are reported from two time points: at the start of the wet season (December 2024), based on data from Structured Community Activity 1 (Community Gathering), the Short Leader Midline Survey, and the first round of Systematic Photographs and Direct Field Observations; and at the peak of the wet season (March 2025), using data from the second round of Systematic Photographs and Direct Field Observations, including the fixed walkability routes.

Tables 6 – 10 present the results in a consistent format. Column (1) reports control group means. Column (2) reports the estimated treatment effect of the intervention, with standard errors shown beneath the coefficient estimates, clustered at the level of randomization. Column (3) indicates the number of observations used in each estimation.

### 5.3.1. *Structured Community Activity 1: Community Gathering*

We begin by assessing whether block leaders organized the meeting after being asked to do so to discuss challenges associated with the approaching rainy season and potential preventive measures. As shown in Table 6, approximately half of the leaders convened their communities, despite all receiving an incentive to support the organization of the gathering. After controlling for block-level covariates, neighborhood fixed effects, and strata fixed effects, we find no evidence that the intervention affected the likelihood of organizing the meeting.

Block leaders were responsible for contacting residents and mobilizing them to attend. On average, leaders reported contacting 29 residents in person, two by phone, and virtually none by SMS. The intervention did not affect the number of residents contacted. These figures are reported only for leaders who organized a meeting, which accounts for the smaller sample size.

On the day of the gathering, enumerators visited each block and recorded the total number of community members present, as well as the number who spoke. On average, 14 residents attended, with about half contributing to the discussion. Enumerators also recorded the names and phone numbers of each attendee, yielding an average of 15, which is consistent with the reported total attendance. The small discrepancy likely reflects residents who arrived after the enumerator had recorded the total but whose details were captured afterward. Again, we find no significant treatment effects.

Finally, block leaders were asked to keep written records of the residents they had contacted, which enumerators collected on the day of the meeting. Counting the names on these lists, we find an average of 30 residents per block. Although lists from the treatment group contain slightly more names, the difference is not statistically significant. Results remain similar if we assign a value of zero to leaders who did not organize a meeting, as reported in Table 6.

Table 7 reports the topics discussed during the community gatherings. On average, four topics were covered. Residents most often discussed garbage disposal and awareness campaigns, and least often mentioned mangrove planting or protection.

After controlling for block-level covariates, neighborhood fixed effects, and strata fixed effects, we find no evidence that the intervention affected either the topics discussed or the number covered. The only exception is that treated communities were significantly less likely to discuss contact with the neighborhood secretary.

A statistic that cannot be compared between treatment and control, because it meaningfully exists only in treated blocks, is the following: in 89 of 126 treated blocks where the leader organized the SCA meeting (71%), the community discussed a topic that matched the implemented activity during the community gathering.

### *5.3.2. Short Leader Midline Survey*

During the first Structured Community Activity, the community gathering described in the previous subsection, we administered a brief survey to block leaders to capture their responses before the rainy season, immediately following implementation of the intervention. Treatment effects from this survey are reported in Table 8.

On average, leaders attended three flood-prevention meetings in the year preceding the survey, and 83% reported speaking with the neighborhood secretary about cleaning drainage or streets. The intervention did not affect either measure. Likewise, 84% of leaders spoke with other block leaders about these issues. Interestingly, leaders in the treatment group were six percentage point less likely to do so, after controlling for block- and individual-level covariates, the corresponding baseline outcome, neighborhood fixed effects, and strata fixed effects. This suggests that the intervention may have reduced the need for coordination with other leaders.

We also asked block leaders to indicate their level of agreement, on a 1–5 scale, with two statements: “It is my own responsibility to prepare my block for an emergency” and “If the community were organized, preventing some flood impacts would be possible.” Agreement in the control group was already high, averaging 4.4 and 4.6, respectively. Nevertheless, leaders in the treatment group were significantly more likely to agree with both statements, by 0.20 and 0.14 points, respectively. These results suggest a shift in attitudes toward personal responsibility and collective efficacy in disaster preparedness, indicating that the intervention

strengthened leaders' sense of agency and their belief in the community's capacity to mitigate flood-related risks through coordinated action.

### 5.3.3. *Systematic Photographs and Direct Field Observations*

**Observation points** We examine impacts at location-specific observation points within blocks: (i) points along drainage canal segments, (ii) block center points, (iii) a low-lying GPS-defined point likely to flood, (iv) household points, and (v) the most visibly flooded point in each block, as identified by the enumerator<sup>8</sup>. Data were collected in December 2024 (start of the wet season) for drainage canal and block center points, and again in March 2025 (peak of the wet season) for all points. Results are presented in Table 9.

**Solid waste.** Enumerators observed visible solid waste at roughly 80–85% of flood-prone, household and most flooded sites in control blocks<sup>9</sup>. The intervention reduced the likelihood of visible solid waste by 2.9 percentage points at household points and 6.7 percentage points at the most flooded points, both significant at the 10% level. No significant treatment effects were found for flood-prone locations. Estimates control for block-level covariates, neighborhood fixed effects, strata fixed effects, and day fixed effects.

To complement the human-coded observations, enumerators photographed all visited locations. For each block, and for each category of points (drainage canal; block center; flood-prone; household; and the most flooded), we calculated the average number of trash detections per photo. For flood-prone points, automated object detection applied to systematic photographs reveals a positive treatment effect on the number of trash detections. No treatment effect was found for other location types (Table 9).

**Dirtiness scores.** Enumerators also assessed the degree of dirtiness at each point on a 0–10 scale, where 0 indicates very clean and 10 indicates very dirty. Drainage canals, flood-prone, and most flooded sites showed no treatment effect

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8. No flooded area was observed in 79 of 489 blocks, accounting for the lower number of observations.

9. Data on solid waste were not collected for drainage canal and block center points.

(with means of 3.4, 3.9, and 4.2). Treated blocks were significantly cleaner at block centers ( $-0.22$ , 5% level) and household points ( $-0.27$ , 5% level), consistent with the intervention’s theory of change.

**Flood preparedness and environmental management.** Enumerators were also asked to assess, at each point, whether there were visible signs of flood preparation measures. Evidence of recent street cleaning increased in treated blocks by 6.9 percentage points at drainage canals sites (10% level; mean 32%) and 4.9 points at the most flooded points (10% level; mean 7%). Recent connections of drainage ditches also increased, by 5 points at drainage canals (10% level; mean 3%), with no effects elsewhere. Signs of *Entulho* or *Abaulamento* rose by 4.1 points at the most flooded points only (5% level; mean 7%). Other measures, such as cleaning of drainage ditches, housing improvements, and bridge construction, showed no significant treatment effects. Slight negative effects were observed at drainage canal sites, which were less likely to exhibit *Entulho/Abaulamento* and housing improvements, by 2 and 4 percentage points, respectively.

**Flooding.** Enumerators reported visible signs of flooding at each observation point. At drainage canal locations, treated blocks showed a modest increase of 5.8 percentage points (over a 27% mean in control blocks), statistically significant. On the other hand, flood-prone sites showed a decrease of 8.3 percentage points (over a 60% mean in control blocks), statistically significant at the 5% level. At block center and household sites the likelihood of visible flooding ranged from 27–51% in control blocks, with no significant treatment effects.

Water levels were measured at flood-prone, household, and most flooded points using reference markers such as house corners or trees. The the mean water level in control blocks ranged from 6.29-20.72 cm, with no significant treatment effect. Data were not collected for drainage canal or block center points.

**Walkability between points** In Table 10, we examine several variables designed to capture walkability and accessibility within blocks and, by extension, the impact of flooding on livelihoods. Enumerators recorded walking times along a predetermined route: a) from the likely flooded area to the first baseline household

house, b) from baseline household house to baseline household house, and c) from the final baseline household house to a block border point next to a tarmac road, yielding five routes. The regressions include block-level covariates, neighborhood fixed effects, strata fixed effects, day fixed effects and enumerator fixed effects.

In control blocks, enumerators reported walking an average of 5 minutes and 10 seconds for each route, measured with a stopwatch. In treated blocks, walking times were 36 seconds shorter, a difference significant at the 1% level. We also examined whether enumerators could follow the most direct path. In 62% of control blocks they could; treatment did not affect this proportion. When deviations occurred, flooding accounted for 7% of deviations in control blocks; treatment reduced this by 3.1 points (5% significance). Other reasons for deviations (physical barriers, impassable roads, security concerns) were unaffected.

Finally, enumerators rated the ease of movement of the route on a scale from 1 (very hard) to 5 (very easy). The average rating in control blocks was 3.24. In treated blocks, enumerators reported greater ease of movement by 0.17 points, significant at the 1% level.

Across multiple measures—solid waste, dirtiness, localized environmental management, flooding prevalence, and walkability—the intervention improved environmental conditions and accessibility. These findings suggest that community-driven initiatives can generate measurable, location-specific impacts on flood preparedness and urban resilience.

#### **5.4. Focus Group Discussions**

The FGDs revealed that residents across the vast majority of blocks evaluated the five implemented measures positively, although challenges were reported in every site (both in successful and unsuccessful blocks). The most frequent problems involved shortages of materials and equipment: six blocks reported insufficient materials (namely, sand and bridge components), and six noted a lack of essential tools such as gloves, boots, and shovels. Weak communication prior to the implemented measures was reported in five blocks, and insufficient technical supervision in four



blocks, particularly for bridge construction and mangrove planting. Three blocks highlighted low-quality materials, and one block faced land access restrictions, which did not allow for the drainage ditch cleaning activity to be completed. Suggested solutions included increasing material and tool provision, improving communication and planning, conducting site visits, and using durable materials and locally suitable species (in the case of mangrove planting).

Residents generally expected continued support, highlighting the need to communicate that assistance is time-bound and resource-constrained. Successful outcomes were closely linked to active engagement by block chiefs, who mobilized volunteers, coordinated in-kind contributions by block residents, and organized volunteer teams. Across all blocks, the block chief was the central coordinator of the implemented measures, and differences in outcomes largely reflected the level of engagement and trust rather than formal organizational structures.

## 6. Conclusion

This paper presents the design, implementation, and preliminary outcomes of a community-driven disaster risk management intervention in the coastal city of Quelimane, Mozambique. Faced with frequent flooding, rapid urbanization, and socioeconomic vulnerability, local communities require not only technical knowledge and financial resources but also mechanisms for collective action and coordination. Our clustered intervention addresses these three key constraints through a combination of technical recommendations, small grants, and structured community meetings that foster cooperation and local ownership.

We find strong community mobilization in treated blocks: during project implementation, more than 4,000 individuals participated as volunteers. The intervention generated modest but meaningful improvements in flood-related behaviors and outcomes. We find shifts in attitudes toward personal responsibility and collective efficacy in disaster preparedness. Treated communities reported higher levels of localized cleaning and more visible flood preparedness. While computer vision-based trash detection showed no consistent reductions, enumerator assessments recorded

improved cleanliness in treated blocks. We also observe less flooded locations and greater ease of mobility in treated areas, suggesting enhanced resilience to seasonal flooding.

While this paper focuses on short-term outcomes, subsequent rounds of data collection will allow us to rigorously assess the remaining outcome families in our theory of change, including demand for leader accountability, changes in knowledge and awareness, and engagement in high-risk behaviors. We will also revisit the outcomes presented in this paper after the rainy season, both to track their evolution and to collect more detailed household- and block-level measures that were not captured in the initial data.

Beyond measuring treatment effects, the project also generates rich spatial data, enabling us to study how risks, responses, and adaptation strategies vary across neighborhoods. Preliminary analyses show that residents within specific neighborhoods exhibit similar willingness to pay for flood mitigation. Further spatial analyses will allow us to address a broad range of research questions.

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## A. Appendix

### A.1. Statistical Power

We calculate the smallest effect size that can be reliably detected in a clustered randomized controlled trial using Stata’s build-in *power* program, including 159 clusters — 79 in control and 80 in treatment — for outcome variables at the household and block leader levels. Each cluster comprises two, three, or four city blocks and, on average, twelve households. The whole experimental sample encompasses 489 blocks, their leaders, and 1,956 households.

We assume a 5% significance level and a 80% statistical power threshold, that is, a 80% probability that our test will yield a statistically significant result if the research hypothesis, in reality, is true. To calculate the smallest detectable effect size, we also measure the intra-cluster correlation coefficients for the different outcome variables based on household and leader baseline survey data. The baseline variables used here are standardized with mean zero and standard deviation one. We treat all clusters as if they had the same size (twelve households).

The results of our power calculations, which show our study’s ability to detect meaningful effects, are depicted in Figures 8 and 9. For household-level outcomes, we show that the experiment is powered to detect treatment effects of 0.17 standard deviations for financial contributions to community-level adaptation efforts. The minimum detectable effect size is 0.22 standard deviations for the likelihood of having been impacted by flooding last year, and 0.25 standard deviations for having seen or participated in collective action. These two overlap in the figure. At the local leader level, we are slightly less powered as there is only one leader per block. Here, the minimum detectable effect size is less than 0.3 standard deviations for the four chosen leader outcome variables.

## A.2. Tables

**Table 1 — Allocation to treatment and control**

|            | Treatment | Control | Total |
|------------|-----------|---------|-------|
| Households | 984       | 972     | 1956  |
| Blocks     | 246       | 243     | 489   |
| Clusters   | 80        | 79      | 159   |

*Note.* Used on page [9] .

**Table 2—Descriptive statistics**

| <b>Panel A. Households</b>           |          |           |        |     |         |
|--------------------------------------|----------|-----------|--------|-----|---------|
|                                      | Mean     | SD        | Median | Min | Max     |
| No. of household members             | 5.36     | 2.33      | 5      | 1   | 19      |
| Female                               | 0.50     | 0.50      | 1      | 0   | 1       |
| Family leader                        | 0.84     | 0.37      | 1      | 0   | 1       |
| Age                                  | 38.97    | 14.53     | 36     | 18  | 97      |
| Worked                               | 0.68     | 0.47      | 1      | 0   | 1       |
| Primary education                    | 0.72     | 0.45      | 1      | 0   | 1       |
| Secondary education                  | 0.40     | 0.49      | 0      | 0   | 1       |
| Monthly income (MZN)                 | 9,365.69 | 11,874.91 | 5,000  | 0   | 200,000 |
| Unmet food needs                     | 0.65     | 0.48      | 1      | 0   | 1       |
| Living with difficulties             | 0.42     | 0.49      | 0      | 0   | 1       |
| Number of flooding episodes          | 1.81     | 2.41      | 1      | 0   | 31      |
| <b>Panel B. Block Leaders</b>        |          |           |        |     |         |
|                                      | Mean     | SD        | Median | Min | Max     |
| Female                               | 0.37     | 0.48      | 0      | 0   | 1       |
| Age                                  | 49.89    | 12.60     | 49     | 20  | 90      |
| Length of position (years)           | 5.55     | 4.82      | 4      | 0   | 40      |
| Worked                               | 0.74     | 0.44      | 1      | 0   | 1       |
| Primary education                    | 0.67     | 0.47      | 1      | 0   | 1       |
| Secondary education                  | 0.19     | 0.40      | 0      | 0   | 1       |
| Household members                    | 6.18     | 2.42      | 6      | 1   | 15      |
| Number of flooding episodes          | 3.00     | 4.21      | 2      | 0   | 50      |
| <b>Panel C. Blocks</b>               |          |           |        |     |         |
|                                      | Mean     | SD        | Median | Min | Max     |
| No leader appointed                  | 0.03     | 0.18      | 0      | 0   | 1       |
| Canals                               | 0.66     | 0.47      | 1      | 0   | 1       |
| Canal type 1 (Covered and paved)     | 0.04     | 0.20      | 0      | 0   | 1       |
| Canal type 2 (Uncovered and paved)   | 0.20     | 0.40      | 0      | 0   | 1       |
| Canal type 3 (Uncovered and unpaved) | 0.42     | 0.49      | 0      | 0   | 1       |
| Area size (h)                        | 6.78     | 8.39      | 4      | 0   | 63      |
| Elevation (m)                        | 7.68     | 1.96      | 8      | 3   | 14      |
| No. of buildings                     | 192.13   | 132.96    | 156    | 32  | 1,139   |

*Note.* Used on page [13] .



**Table 3 — Balance tests: Households**

| Variable                                                                         | (1)<br>Control |                   | (2)<br>Treatment |                   | (1)-(2)<br>Pairwise t-test |                 |
|----------------------------------------------------------------------------------|----------------|-------------------|------------------|-------------------|----------------------------|-----------------|
|                                                                                  | N/Clusters     | Mean/(SE)         | N/Clusters       | Mean/(SE)         | N/Clusters                 | Mean difference |
| Household: Female                                                                | 972<br>79      | 0.500<br>(0.000)  | 984<br>80        | 0.500<br>(0.000)  | 1956<br>159                | 0.000           |
| Household: Age                                                                   | 972<br>79      | 39.393<br>(0.544) | 984<br>80        | 38.556<br>(0.595) | 1956<br>159                | 0.837           |
| Household: Worked                                                                | 972<br>79      | 0.675<br>(0.022)  | 984<br>80        | 0.679<br>(0.021)  | 1956<br>159                | -0.004          |
| Household: Knows chief's name (H1)                                               | 972<br>79      | 0.483<br>(0.025)  | 984<br>80        | 0.473<br>(0.030)  | 1956<br>159                | 0.010           |
| Household: Knows drainage should be kept trash-free (H2)                         | 972<br>79      | 0.761<br>(0.020)  | 983<br>80        | 0.737<br>(0.021)  | 1955<br>159                | 0.025           |
| Household: Works with community to achieve common goals (H3)                     | 968<br>79      | 0.264<br>(0.021)  | 983<br>80        | 0.268<br>(0.020)  | 1951<br>159                | -0.003          |
| Household: Strongly believes preparedness reduces flood risk (H4)                | 971<br>79      | 0.015<br>(0.005)  | 984<br>80        | 0.010<br>(0.003)  | 1955<br>159                | 0.005           |
| Household: Participated in a cleaning task (H6)                                  | 969<br>79      | 0.543<br>(0.024)  | 979<br>80        | 0.541<br>(0.025)  | 1948<br>159                | 0.001           |
| Household: Collected preparedness information (H7)                               | 972<br>79      | 0.599<br>(0.026)  | 979<br>80        | 0.623<br>(0.028)  | 1951<br>159                | -0.024          |
| Household: Block has constructions that block pathways (H8)                      | 964<br>79      | 0.481<br>(0.029)  | 971<br>80        | 0.522<br>(0.031)  | 1935<br>159                | -0.041          |
| Household: Water level inside house during last rainy season (cm) (H9)           | 968<br>79      | 17.771<br>(2.166) | 981<br>80        | 17.951<br>(2.620) | 1949<br>159                | -0.180          |
| Household: Impacts of last rainy season greater than median monthly income (H10) | 952<br>79      | 0.800<br>(0.020)  | 964<br>80        | 0.771<br>(0.022)  | 1916<br>159                | 0.030           |

*Note:* Significance: \*\* 0.01, \* 0.05, + 0.1. Standard errors (reported in parentheses) are clustered at the level of randomization. The regression includes strata fixed effects. The values displayed for t-tests are the differences in means across groups. Used on page [13] .

**Table 4 — Balance tests: Blocks**

| Variable                                                                                                 | (1)<br>Control |                     | (2)<br>Treatment |                     | (1)-(2)<br>Pairwise t-test |                    |
|----------------------------------------------------------------------------------------------------------|----------------|---------------------|------------------|---------------------|----------------------------|--------------------|
|                                                                                                          | N/Clusters     | Mean/(SE)           | N/Clusters       | Mean/(SE)           | N/Clusters                 | Mean difference    |
| Canals                                                                                                   | 243<br>79      | 0.675<br>(0.044)    | 246<br>80        | 0.654<br>(0.042)    | 489<br>159                 | 0.020              |
| Canal type 1 (Covered and paved)                                                                         | 243<br>79      | 0.045<br>(0.017)    | 246<br>80        | 0.041<br>(0.021)    | 489<br>159                 | 0.005              |
| Canal type 2 (Uncovered and paved)                                                                       | 243<br>79      | 0.214<br>(0.041)    | 246<br>80        | 0.191<br>(0.036)    | 489<br>159                 | 0.023              |
| Canal type 3 (Uncovered and unpaved)                                                                     | 243<br>79      | 0.416<br>(0.048)    | 246<br>80        | 0.423<br>(0.045)    | 489<br>159                 | -0.007             |
| Area size (h)                                                                                            | 243<br>79      | 7.354<br>(0.896)    | 246<br>80        | 6.210<br>(0.653)    | 489<br>159                 | 1.144              |
| Elevation (m)                                                                                            | 243<br>79      | 7.772<br>(0.195)    | 246<br>80        | 7.580<br>(0.190)    | 489<br>159                 | 0.192              |
| No. of buildings                                                                                         | 243<br>79      | 191.169<br>(11.358) | 246<br>80        | 193.081<br>(11.201) | 489<br>159                 | -1.913             |
| Chief: Female                                                                                            | 243<br>79      | 0.362<br>(0.037)    | 246<br>80        | 0.370<br>(0.032)    | 489<br>159                 | -0.008             |
| Chief: Age                                                                                               | 243<br>79      | 49.826<br>(0.977)   | 246<br>80        | 49.955<br>(0.988)   | 489<br>159                 | -0.129             |
| Chief: Worked                                                                                            | 243<br>79      | 0.745<br>(0.028)    | 246<br>80        | 0.740<br>(0.033)    | 489<br>159                 | 0.005              |
| Chief: Knows drainage should be kept trash-free (H2)                                                     | 243<br>79      | 0.835<br>(0.022)    | 246<br>80        | 0.797<br>(0.026)    | 489<br>159                 | 0.039              |
| Chief: Works with community to achieve common goals (H3)                                                 | 243<br>79      | 0.679<br>(0.039)    | 246<br>80        | 0.687<br>(0.033)    | 489<br>159                 | -0.008             |
| Chief: Agrees with "If the community was organized, preventing some flood impact would be possible" (H4) | 243<br>79      | 0.012<br>(0.009)    | 246<br>80        | 0.008<br>(0.006)    | 489<br>159                 | 0.004              |
| Chief: Number of meetings attended in last 12 months (H5)                                                | 238<br>79      | 11.101<br>(0.796)   | 238<br>80        | 9.298<br>(0.798)    | 476<br>159                 | 1.803 <sup>+</sup> |
| Chief: Saw community members participate in a cleaning task (H6)                                         | 242<br>79      | 0.872<br>(0.025)    | 246<br>80        | 0.870<br>(0.025)    | 488<br>159                 | 0.002              |
| Chief: Collected preparedness information (H7)                                                           | 241<br>79      | 0.776<br>(0.029)    | 245<br>80        | 0.776<br>(0.030)    | 486<br>159                 | 0.000              |
| Chief: Block has constructions that block pathways (H8)                                                  | 243<br>79      | 0.560<br>(0.038)    | 246<br>80        | 0.602<br>(0.036)    | 489<br>159                 | -0.042             |
| Chief: Block has flooding experience (H9)                                                                | 243<br>79      | 0.765<br>(0.031)    | 246<br>80        | 0.776<br>(0.029)    | 489<br>159                 | -0.011             |
| Chief: Quality of life decreased after rainy season (H10)                                                | 243<br>79      | 0.786<br>(0.030)    | 245<br>80        | 0.747<br>(0.033)    | 488<br>159                 | 0.039              |

*Note:* Significance: \*\* 0.01, \* 0.05, + 0.1. Standard errors (reported in parentheses) are clustered at the level of randomization. The regression includes strata fixed effects. The values displayed for t-tests are the differences in means across groups. The block variables are defined by the following: 1. Area size in hectares: variable calculated using the mapped borders in QGIS; 2. Elevation in meters: variable capturing the average elevation in meters calculated using SRTM data V4 and the mapped borders in QGIS; 3. Number of buildings: variable calculated using OpenStreetMap and the mapped borders in QGIS. Used on page [13] .

Table 5 — Spatial clustering of willingness to pay for flood mitigation

|                              | Mor. I    | Mean WTP |
|------------------------------|-----------|----------|
| <i>Near Coastline</i>        | 0.0412*** | 3.08     |
| <i>Not Near Coastline</i>    | 0.0329*** | 3.04     |
| <i>Mangrove Adjacent</i>     | 0.0786*** | 2.65     |
| <i>Not Mangrove Adjacent</i> | 0.0236*** | 3.13     |
| <i>High Pop. Density</i>     | 0.0242*** | 3.13     |
| <i>Low Pop. Density</i>      | 0.0453*** | 2.98     |
| <i>High Elevation</i>        | 0.021***  | 3.20     |
| <i>Low Elevation</i>         | 0.0441*** | 2.92     |

**Table 6 — Treatment effects: Community gathering (SCA 1)**

|                                                                                          | Control<br>Mean  | TE              | Obs. |
|------------------------------------------------------------------------------------------|------------------|-----------------|------|
| Leader organized the meeting                                                             | 0.527<br>0.500   | -0.017<br>0.040 | 487  |
| No. of residents contacted in person (reported by leader)                                | 28.961<br>30.741 | 2.736<br>2.902  | 253  |
| No. of residents contacted by phone (reported by leader)                                 | 1.781<br>6.337   | -0.170<br>0.692 | 253  |
| No. of residents contacted by SMS (reported by leader)                                   | 0.383<br>2.137   | -0.007<br>0.166 | 253  |
| No. of residents present (reported by enumerator)                                        | 14.102<br>10.584 | 2.143<br>1.400  | 253  |
| No. of residents speaking (reported by enumerator)                                       | 6.893<br>5.669   | 0.110<br>0.674  | 247  |
| No. of recorded names of residents present                                               | 14.655<br>11.015 | 1.865<br>1.418  | 227  |
| No. of recorded phone numbers of residents present                                       | 14.655<br>11.015 | 1.865<br>1.418  | 227  |
| No. of residents contacted in total (on paper)                                           | 29.871<br>33.992 | 3.641<br>3.877  | 235  |
| No. of residents contacted in total (on paper)<br>(0 if leader did not organize meeting) | 14.281<br>27.680 | 2.218<br>2.160  | 450  |

*Note.* The regression includes the number of houses, area size (in hectares), length of drainage canals (in meters), average elevation (in meters), neighborhood fixed effects, and strata fixed effects as control variables at block level. Column (1) reports control group means at midline and standard deviations below; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization with the original form of the sandwich estimator (Liang and Zeger, 1986); column (3) reports the No. of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels. Used on page 23.

**Table 7 — Treatment effects: Topics discussed during community gathering (SCA 1)**

|                                                                               | Control<br>Mean | TE               | Obs. |
|-------------------------------------------------------------------------------|-----------------|------------------|------|
| No. of topics discussed during the community meeting                          | 3.836<br>1.451  | -0.287<br>0.207  | 253  |
| Community discussed <b>garbage disposal</b> during meeting                    | 0.758<br>0.430  | 0.058<br>0.049   | 253  |
| Community discussed <b>unauthorized construction</b> during meeting           | 0.352<br>0.479  | -0.076<br>0.064  | 253  |
| Community discussed <b>Entulho or Abaulamento</b> during meeting              | 0.492<br>0.502  | -0.051<br>0.064  | 253  |
| Community discussed <b>bridge construction</b> during meeting                 | 0.125<br>0.332  | -0.004<br>0.041  | 253  |
| Community discussed <b>mangrove planting/protection</b> during meeting        | 0.117<br>0.323  | 0.013<br>0.035   | 253  |
| Community discussed <b>planting vegetation around houses</b> during meeting   | 0.156<br>0.365  | -0.042<br>0.037  | 253  |
| Community discussed <b>distribution of flood warnings</b> during meeting      | 0.438<br>0.498  | 0.025<br>0.067   | 253  |
| Community discussed <b>community collaboration</b> during meeting             | 0.570<br>0.497  | -0.034<br>0.056  | 253  |
| Community discussed <b>contact with neighborhood secretary</b> during meeting | 0.195<br>0.398  | -0.097*<br>0.042 | 253  |
| Community discussed <b>awareness campaigns</b> during meeting                 | 0.633<br>0.484  | -0.080<br>0.050  | 253  |

*Note.* The regression includes the number of houses, area size (in hectares), length of drainage canals (in meters), average elevation (in meters), neighborhood fixed effects, and strata fixed effects as control variables at block level. Column (1) reports control group means at midline and standard deviations below; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization with the original form of the sandwich estimator (Liang and Zeger, 1986); column (3) reports the No. of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels. Used on page [23] .

**Table 8 — Treatment effects: Short midline leader survey**

|                                                                                                    | Control<br>Mean | TE               | Obs. |
|----------------------------------------------------------------------------------------------------|-----------------|------------------|------|
| No. of flood-prevention meetings leader attended last year                                         | 2.527<br>2.765  | 0.222<br>0.277   | 463  |
| Leader spoke with neighborhood secretary about cleaning drainage/streets                           | 0.831<br>0.375  | -0.053<br>0.034  | 483  |
| Leader spoke with other block leaders about cleaning drainage/streets                              | 0.844<br>0.364  | -0.061*<br>0.026 | 484  |
| Agrees with: It is my own responsibility to prepare my block for an emergency. (1-5)               | 4.374<br>0.909  | 0.202*<br>0.079  | 476  |
| Agrees with: If the community was organized, preventing some flood impact would be possible. (1-5) | 4.600<br>0.713  | 0.139*<br>0.058  | 479  |

*Note.* The regression includes the number of houses, area size (in hectares), length of drainage canals (in meters), average elevation (in meters), neighborhood fixed effects, and strata fixed effects as control variables at block level. In addition, it includes the respondent’s gender, age, household size, and an indicator variable equal to 1 if they have some secondary education and 0 otherwise, as individual-level control variables. The corresponding outcome at baseline is also included as a control variable. Variables are not transformed.

Column (1) reports control group means at midline and standard deviations below; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization with the original form of the sandwich estimator (Liang and Zeger, 1986), which does not make any small-sample correction; column (3) reports the No. of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels. Used on page [24] .

Table 9 — Treatment effects: All observed locations

|                                                                                                                    | Drainage canal points |                  |      | Block center points |                  |      | Flood-prone points |                  |      | Household points |                  |      | Most flooded point |                  |      |
|--------------------------------------------------------------------------------------------------------------------|-----------------------|------------------|------|---------------------|------------------|------|--------------------|------------------|------|------------------|------------------|------|--------------------|------------------|------|
|                                                                                                                    | Control Mean          | TE               | Obs. | Control Mean        | TE               | Obs. | Control Mean       | TE               | Obs. | Control Mean     | TE               | Obs. | Control Mean       | TE               | Obs. |
| Enumerator reported solid waste in sight<br>(0 = no, 1 = yes)                                                      | —                     | —                | —    | —                   | —                | —    | 0.798              | -0.011<br>0.030  | 489  | 0.819            | -0.029+<br>0.016 | 1956 | 0.846              | -0.067+<br>0.038 | 416  |
| No. of trash detections at or around points                                                                        | 3.902                 | -0.013<br>0.364  | 2103 | 4.806               | -0.128<br>0.310  | 968  | 6.175              | 1.341**<br>0.461 | 489  | 7.585            | -0.281<br>0.344  | 1956 | 5.107              | 0.574<br>0.408   | 416  |
| Enumerator-reported dirtiness of the location<br>(scale of 0 to 10: 0 = very clean, 10 = very dirty)               | 3.373                 | -0.140<br>0.248  | 2103 | 3.539               | -0.215*<br>0.101 | 968  | 3.852              | -0.244<br>0.182  | 489  | 3.730            | -0.273*<br>0.120 | 1956 | 4.154              | -0.168<br>0.195  | 416  |
| Enumerator reported block residents recently <b>cleaning streets</b><br>(0 = no, 1 = yes)                          | 0.316                 | 0.069+<br>0.039  | 2103 | 0.293               | -0.040<br>0.026  | 968  | 0.169              | 0.033<br>0.028   | 489  | 0.193            | 0.015<br>0.024   | 1956 | 0.067              | 0.049+<br>0.026  | 416  |
| Enumerator-reported block residents recently <b>connecting ditches to the drainage system</b><br>(0 = no, 1 = yes) | 0.030                 | 0.050+<br>0.028  | 2103 | 0.048               | -0.011<br>0.015  | 968  | 0.021              | 0.014<br>0.013   | 489  | 0.036            | -0.008<br>0.011  | 1956 | 0.038              | -0.010<br>0.023  | 416  |
| Enumerator-reported block residents recently <b>cleaning drainage ditches</b><br>(0 = no, 1 = yes)                 | —                     | —                | —    | —                   | —                | —    | 0.021              | -0.005<br>0.013  | 489  | 0.019            | -0.006<br>0.010  | 1956 | 0.005              | -0.003<br>0.005  | 416  |
| Enumerator-reported recent <i>Entulho</i> or <i>Abaulamento</i> by block residents<br>(0 = no, 1 = yes)            | 0.017                 | -0.021*<br>0.009 | 2103 | 0.046               | 0.018<br>0.013   | 968  | 0.095              | 0.015<br>0.022   | 489  | 0.080            | -0.012<br>0.013  | 1956 | 0.067              | 0.041*<br>0.019  | 416  |
| Enumerator-reported block residents recently <b>making housing improvements</b><br>(0 = no, 1 = yes)               | 0.061                 | -0.039+<br>0.020 | 2103 | 0.071               | -0.011<br>0.012  | 968  | 0.206              | -0.051<br>0.032  | 489  | 0.212            | 0.080<br>0.026   | 1956 | 0.188              | 0.028<br>0.039   | 416  |
| Enumerator-reported block residents recently <b>constructing bridges</b><br>(0 = no, 1 = yes)                      | 0.002                 | 0.005<br>0.004   | 2103 | 0.006               | -0.002<br>0.006  | 968  | 0.004              | -0.005<br>0.004  | 489  | 0.008            | 0.001<br>0.003   | 1956 | 0.010              | 0.005<br>0.005   | 416  |
| Enumerator reported flooded areas in sight<br>(0 = no, 1 = yes)                                                    | 0.269                 | 0.058*<br>0.029  | 2103 | 0.268               | 0.013<br>0.023   | 968  | 0.597              | -0.083*<br>0.039 | 489  | 0.511            | 0.008<br>0.028   | 1956 | —                  | —                | —    |
| Enumerator-reported water depth at the most flooded point in sight<br>(in centimeters)                             | —                     | —                | —    | —                   | —                | —    | 9.275              | -1.430<br>0.956  | 489  | 6.292            | -0.854<br>0.581  | 1956 | 20.918             | -1.724<br>1.280  | 416  |

*Note:* The drainage canal points and the block center points regressions include the number of houses, area size (in hectares), length of drainage canals (in meters), average elevation (in meters), neighborhood fixed effects, and strata fixed effects as control variables at block level. They do also include an indicator for the data collection round (December 2024 and March 2025). The flood-prone points, household points, and most flooded points regressions include the number of houses, area size (in hectares), length of drainage canals (in meters), average elevation (in meters), neighborhood fixed effects, strata fixed effects, and day fixed effects as control variables at block level. The variable «No. of trash detections at or around points» was winsorized at the bottom and top 1% of its distribution. The variable «Enumerator-reported water depth at the most flooded point in sight» was winsorized at the top 1% of its distribution.

Columns (1), (4), (7), (9), and (11) report control group means; columns (2), (5), (8), (11), and (14) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization (with the original form of the sandwich estimator proposed by (Liang and Zeger, 1986)); columns (3), (6), (9), (12), and (15) report the No. of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels. In December 2025, enumerators visited: (1) points every 50 meters along drainage canals and (2) the central point within each block. In March 2025, enumerators visited these same points and also: (1) one low-elevation point identified as flood-prone, (2) four points located in front of baseline household houses, and (3) the most flooded point in each block, as identified by the enumerators. Because flooding was reported to have occurred in only 410 of the 489 blocks, the most flooded point in each block could not be identified in the remaining 79 blocks. Used on page 25.

Table 10 — Treatment effects: Walkability between points

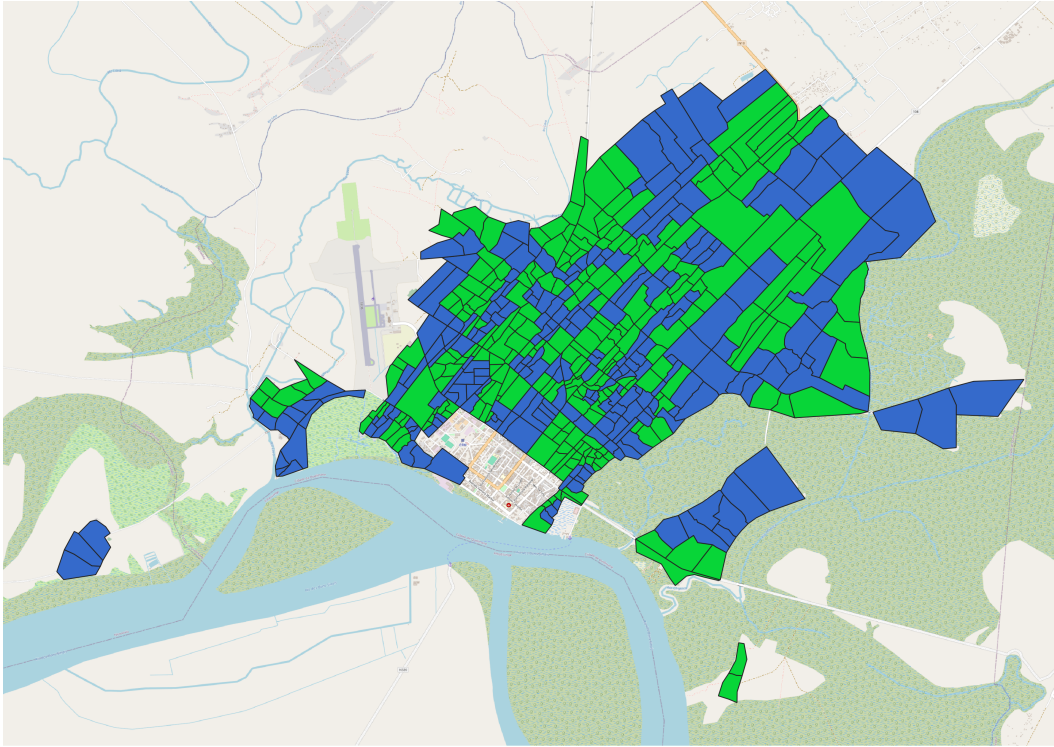
|                                                                                                          | Control<br>Mean | TE                  | Obs. |
|----------------------------------------------------------------------------------------------------------|-----------------|---------------------|------|
| Enumerator-reported walking time between two points<br>(in seconds, timed with a stopwatch)              | 310.453         | -35.668**<br>12.599 | 2445 |
| Enumerator followed the most direct path between two points<br>(0 = no, 1 = yes)                         | 0.622           | 0.025<br>0.022      | 2445 |
| Enumerator deviated between two points <b>due to flooding</b><br>(0 = no, 1 = yes)                       | 0.074           | -0.031*<br>0.013    | 2445 |
| Enumerator deviated between two points <b>due to physical barriers</b><br>(0 = no, 1 = yes)              | 0.235           | 0.000<br>0.019      | 2445 |
| Enumerator deviated between two points <b>due to an impassable road</b><br>(0 = no, 1 = yes)             | 0.138           | -0.010<br>0.017     | 2445 |
| Enumerator deviated between two points <b>due to security concerns</b><br>(0 = no, 1 = yes)              | 0.028           | -0.008<br>0.006     | 2445 |
| Enumerator-reported ease of moving between two points<br>(scale of 1 to 5: 0 = very hard, 5 = very easy) | 3.239           | 0.166**<br>0.055    | 2445 |

*Note:* The regression includes the number of houses, area size (in hectares), length of drainage canals (in meters), average elevation (in meters), neighborhood fixed effects, strata fixed effects, day fixed effects, and enumerator fixed effects as control variables at block level. Column (1) reports control group means; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization (with the original form of the sandwich estimator proposed by (Liang and Zeger, 1986)); column (3) reports the No. of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels. In March 2025, enumerators walked along a predetermined route: (1) from the flood-prone point to the first baseline household house, (2) from baseline household house to baseline household house, and (3) from the final baseline household house to a border point by a tarmac road. Used on pages [22,26] .



### A.3. Figures

**Figure 1 — Blocks by treatment group**



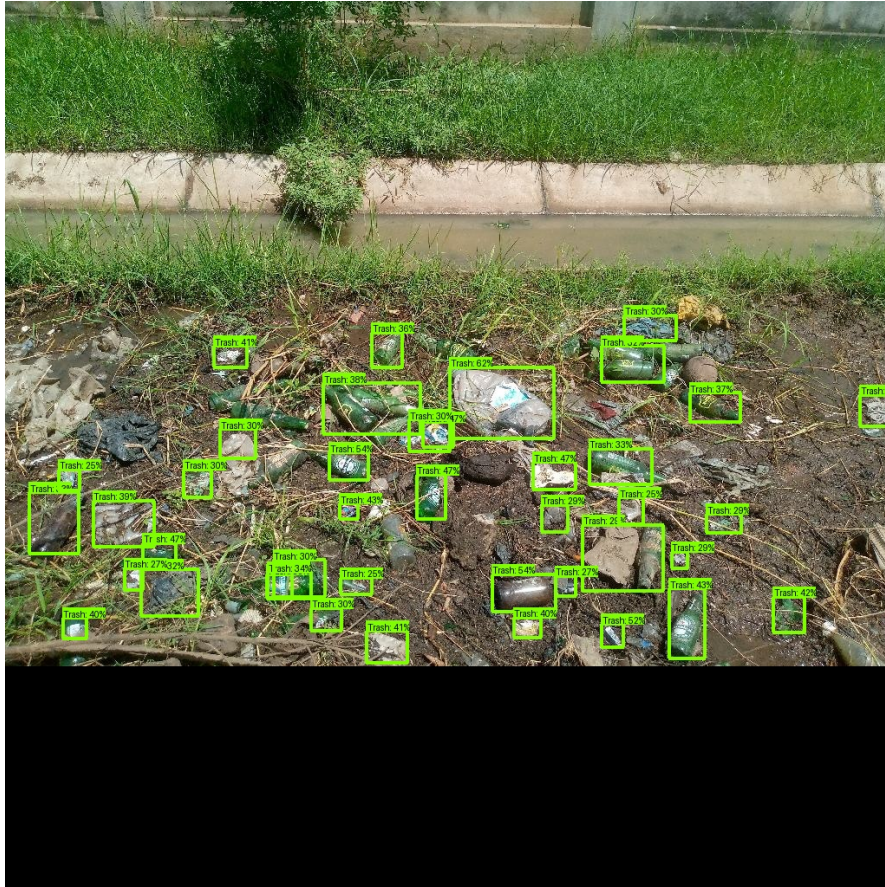
*Note.* Treatment blocks in green. Basemap: © OpenStreetMap contributors. Used on page [9] .

Figure 2 — Theory of change



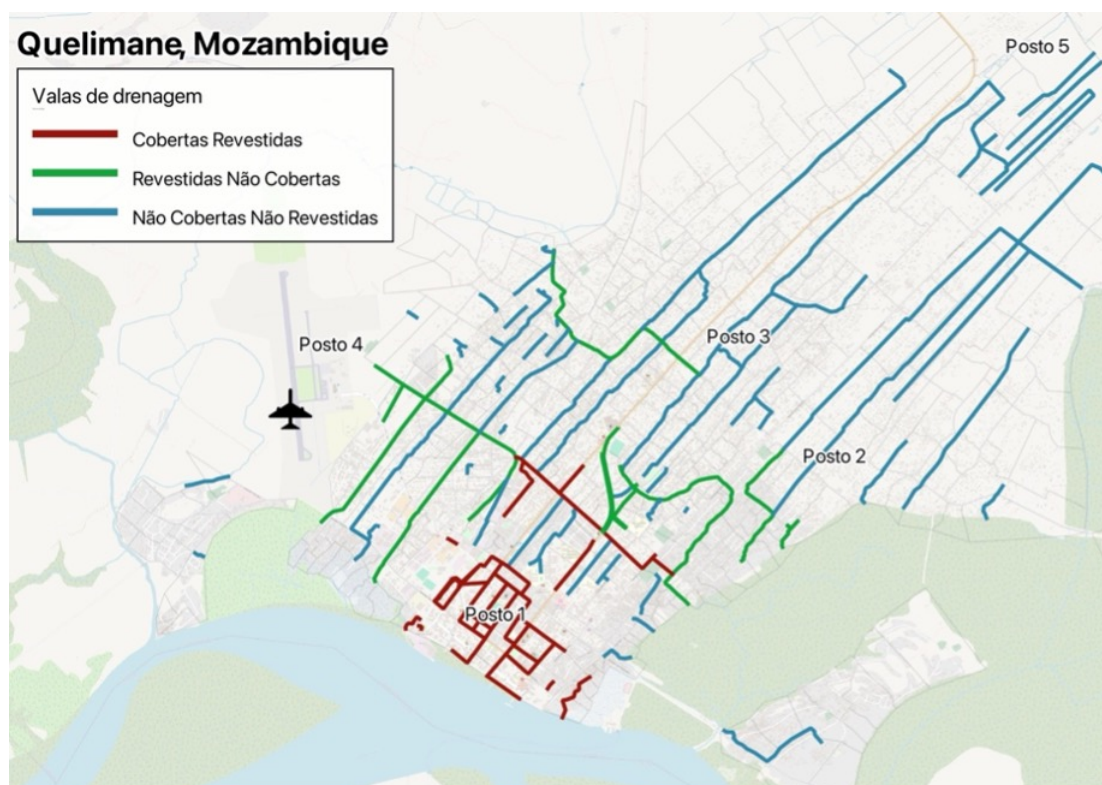
Note. Used on page [9,9] .

Figure 3 — Example of trash detection predictions



*Note.* Used on page [16] .

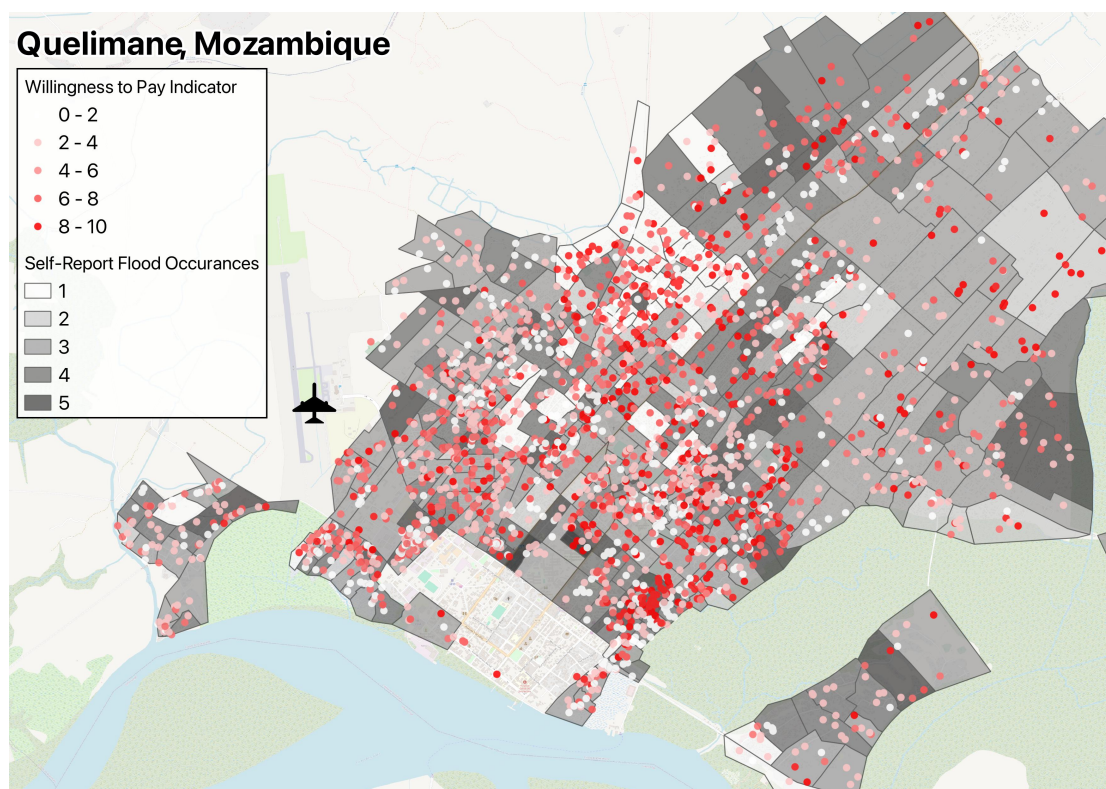
Figure 4 — Map of drainage system



*Note.* Used on page [20] .



Figure 5 — Map of flood occurrence and willingness to pay



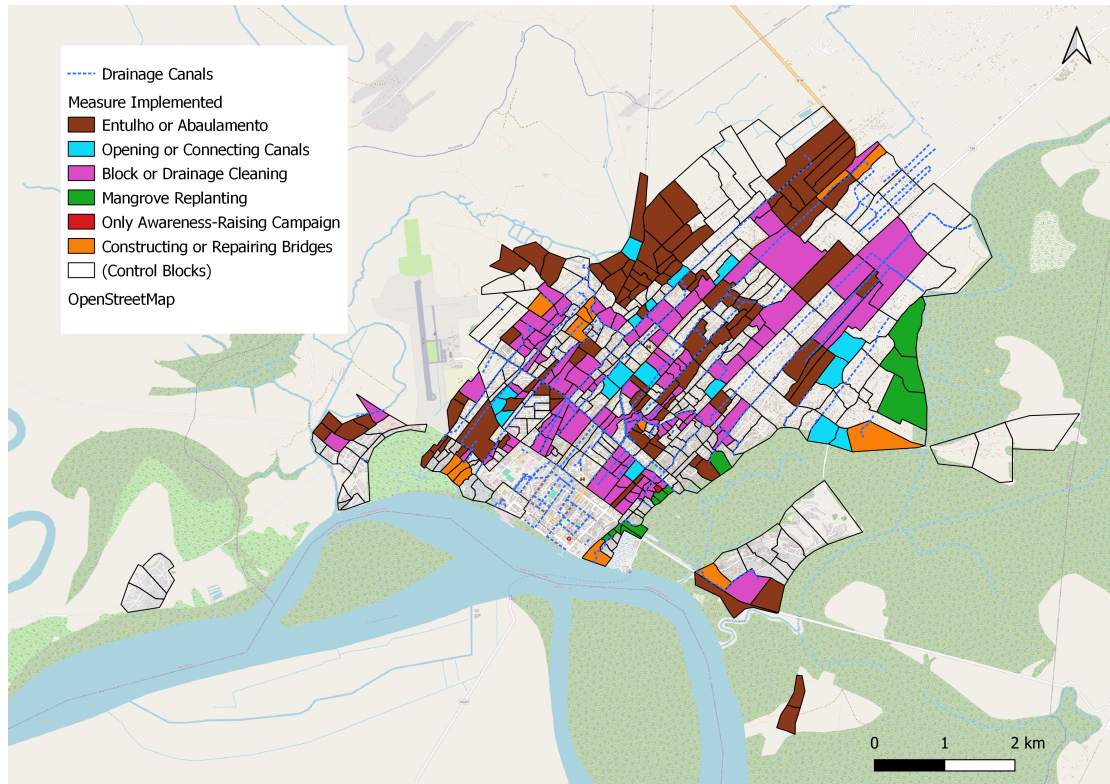
*Note.* Used on page [19] .

Figure 6 — Map of areas proposed for mangrove replanting



*Note.* Used on page [20] .

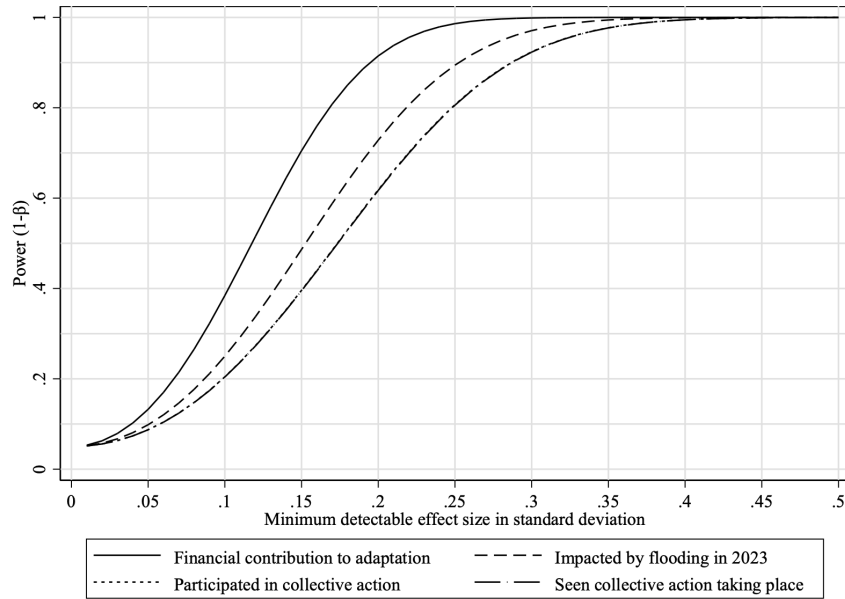
**Figure 7 — Map of implemented measures**



*Note.* Used on page [21] .

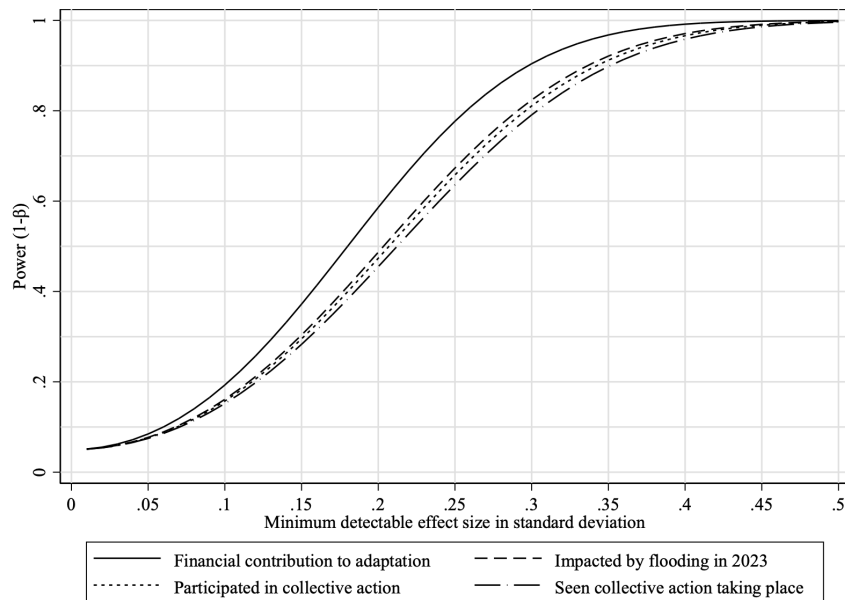


**Figure 8 — Statistical power calculations (households)**



*Note.* Significance level is 0.05, both groups have 79 clusters and the sample standard deviation is 1. The average cluster size is 12. The intra-cluster correlation varies by outcome variable and is calculated using the baseline dataset. Used on page [33] .

**Figure 9 — Statistical power calculations (block leaders)**



*Note.* Significance level is 0.05, both groups have 79 clusters and the sample standard deviation is 1. The average cluster size is 3. The intra-cluster correlation varies by outcome variable and is calculated using the baseline dataset. Used on page [33] .